

Case Study Report: Hotel Booking Demand Analysis Using Exploratory Data Analysis (EDA)

1. Executive Summary

The hospitality industry relies heavily on data-driven decision-making to optimize revenue, improve customer satisfaction, and reduce booking cancellations. This project analyzes hotel booking data to uncover patterns related to customer behavior, room preferences, booking trends, pricing strategies, and cancellation behavior.

Using Python and Exploratory Data Analysis (EDA) techniques, the dataset was cleaned, transformed, and analyzed through descriptive statistics, visualizations, correlation analysis, and predictive modeling. The study provides actionable insights that can help hotel management improve operational efficiency and profitability.

Key findings reveal that room type A dominates bookings, previous cancellations strongly influence future cancellations, family bookings tend to generate higher ADR (Average Daily Rate), and booking cancellations exhibit seasonal patterns.

2. Project Objective

The primary objectives of this project were:

- Understand hotel booking behavior.
 - Identify factors influencing booking cancellations.
 - Analyze customer demographics and their impact on pricing.
 - Examine seasonal booking and cancellation trends.
 - Evaluate room demand and market segment behavior.
 - Generate actionable business insights for hotel management.
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3. Dataset Overview

The dataset contains hotel reservation information including:

Customer Information

- Adults
- Children
- Babies
- Repeated Guest Status

Booking Information

- Lead Time
- Booking Changes
- Previous Cancellations
- Special Requests

Revenue Information

- ADR (Average Daily Rate)

Reservation Information

- Reserved Room Type
- Market Segment
- Arrival Month

Target Variable

- is_canceled

The dataset consists of thousands of booking records representing customer reservation behavior.

4. Data Cleaning and Preparation

Before analysis, several preprocessing steps were performed:

Data Cleaning Activities

- Removed duplicate records.
- Checked and handled missing values.
- Corrected data types where necessary.
- Created new feature:
 - $\text{Total Stay} = \text{Week Nights} + \text{Weekend Nights}$
- Removed negative ADR values.
- Standardized categorical variables.
- Verified consistency across booking information.

Outcome

The cleaned dataset (`df_cleaned`) was used for all subsequent analyses.

5. Exploratory Data Analysis (Basic & Intermediate Findings)

5.1 Impact of Special Requests on ADR

Analysis showed that ADR generally increases as the number of special requests increases.

Special Requests Average ADR

0	99.80
1	109.78
2	118.70
3	125.23
4	131.09
5	129.98

Insight

Guests making more special requests tend to pay higher room rates.

Business Recommendation

Premium guests can be identified through special request behavior and targeted with personalized services.

5.2 Average Stay Duration: New vs Repeated Guests

Guest Type	Average Stay (Days)
New Guests	3.70
Repeated Guests	1.95

Insight

New guests stay almost twice as long as repeated guests.

Business Recommendation

Hotels should encourage repeat guests to extend their stay through loyalty rewards and special offers.

5.3 Most Popular Room Type

Room Type	Bookings
A	56,417

Room Type A accounts for:

64.69% of Total Bookings

Insight

Room Type A is the most demanded room category.

Business Recommendation

Inventory planning should prioritize Room Type A availability.

6. Advanced Analysis

6.1 Factors Affecting Cancellation Rate

Method

A Logistic Regression model was developed using:

- Lead Time
- Previous Cancellations
- Booking Changes
- Total Special Requests
- ADR

Model Accuracy

73.79%

Feature Importance

Feature	Coefficient
Previous Cancellations	0.605
ADR	0.007
Lead Time	0.005
Special Requests	-0.461
Booking Changes	-0.497

Findings

- Previous cancellations are the strongest predictor of future cancellations.
- Guests with more booking changes tend to cancel less.
- Guests making special requests are less likely to cancel.

Business Impact

Hotels can proactively identify high-risk reservations and implement retention strategies.

6.2 Impact of Guest Composition on ADR

Method

Multiple Linear Regression was used with:

- Adults

- Children
- Babies

Predicting:

- ADR

Model Performance

R² Score: **0.214**

Regression Coefficients

Feature Coefficient

Children 38.39

Adults 32.86

Babies 7.61

Findings

Children have the strongest influence on room pricing.

Business Impact

Family-oriented pricing packages can improve revenue generation.

6.3 Impact of Booking Changes on Guest Satisfaction

Special requests were used as an indirect indicator of guest satisfaction.

Results

Correlation:

0.018

Regression Coefficient:

0.021

R² Score:

0.0003

Findings

There is virtually no meaningful relationship between booking changes and special requests.

Business Impact

Hotels should focus on other satisfaction indicators rather than booking modifications.

6.4 Seasonal Impact on Booking Cancellations

Cancellation Rate by Month

Month	Cancellation Rate (%)
January	22.14
February	23.22
March	24.42
April	30.46
May	29.27
June	30.34
July	31.83
August	32.22
September	24.47
October	23.68
November	21.16
December	26.94

Highest Cancellation Month

August (32.22%)

Lowest Cancellation Month

November (21.16%)

Business Impact

Hotels should adopt stricter cancellation policies during peak cancellation months.

6.5 Lead Time Distribution Across Market Segments

Average Lead Time

Market Segment Lead Time (Days)

Groups	147.91
Offline TA/TO	106.08
Online TA	79.97
Direct	48.69
Corporate	16.27
Complementary	13.82
Aviation	4.30
Undefined	1.50

Findings

Group bookings are planned significantly earlier than other segments.

Business Impact

Hotels can offer early-bird discounts and promotions to group customers.

7. Key Business Insights

Revenue Management

- Higher special requests are associated with higher ADR.
- Family bookings contribute significantly to revenue.

Customer Retention

- Previous cancellation history strongly predicts future cancellations.
- Repeat guests have shorter stay durations.

Inventory Optimization

- Room Type A should receive priority allocation due to high demand.

Seasonal Planning

- August experiences the highest cancellation rate.
- November experiences the lowest cancellation rate.

Market Segmentation

- Group customers book much earlier than other customer segments.
 - Corporate and aviation customers often make last-minute reservations.
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8. Recommendations

Recommendation 1

Develop targeted retention campaigns for customers with prior cancellation histories.

Recommendation 2

Offer family-focused packages and promotions to increase ADR.

Recommendation 3

Prioritize Room Type A inventory planning.

Recommendation 4

Implement seasonal cancellation policies during high-risk months.

Recommendation 5

Create early booking incentives for group reservations.

Recommendation 6

Use predictive analytics models to identify cancellation risks before arrival.

9. Additional Considerations

Ethical and Privacy Considerations

- Customer information should be protected and handled responsibly.
- Data privacy regulations must be followed.

Cross-Functional Collaboration

- Insights should be validated with revenue managers, marketing teams, and operational staff.

Continuous Improvement

- Models should be retrained regularly using updated booking data.
 - Business strategies should evolve as customer behavior changes.
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10. Conclusion

This project successfully analyzed hotel booking demand data using Exploratory Data Analysis and predictive modeling techniques. The analysis identified important relationships between booking behavior, pricing, cancellations, room demand, and market segments. The findings demonstrate that previous cancellations are the strongest indicator of future cancellations, family bookings contribute positively to ADR, and seasonal patterns significantly influence cancellation behavior.

The insights generated from this study can support hotel management in optimizing pricing strategies, reducing cancellations, improving customer retention, and enhancing operational efficiency. By leveraging these findings, hotels can make more informed business decisions and achieve sustainable revenue growth.